

DR3500CA Rear Helper Spring Kit Installation Manual

Model: DR3500CA (Rear Suspension Package)

Please read these instructions carefully before starting the installation. Ensure all standard safety precautions are observed while working underneath the vehicle.

1. Parts List

Before beginning, verify that all components listed below are present in your kit.

Ref #	Description / Specifications	Qty
1	Weldment Bracket (Left-Hand & Right-Hand)	1L - 1R
2	Rubber Spacer (4" O.D. x 1" Thick with AH Bushing)	2
3	Hollow Rubber Spring	2
4	Hex Cap Screw (HCS) 3/8" x 1-3/4" UNC Gr 5 Z	2
5	Hex Cap Screw (HCS) 3/8" x 2-3/4" UNC Gr 5 Z	2

Assembly / Installation Diagram

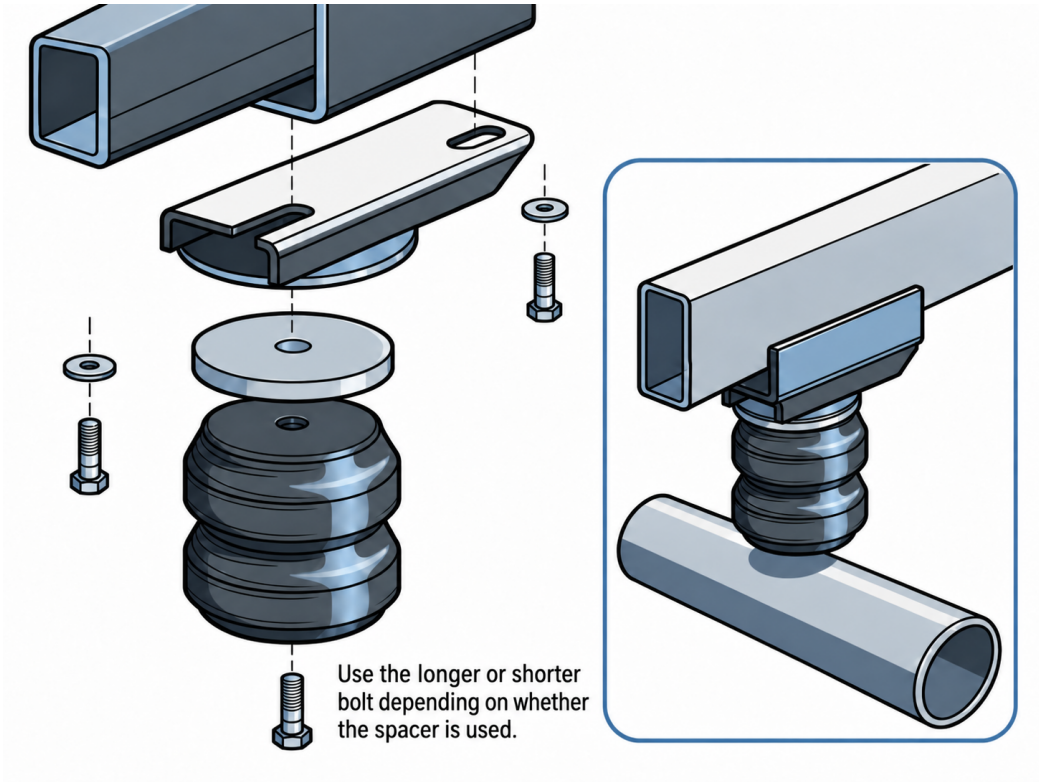


Illustration is for installation reference. Use the longer or shorter bolt depending on whether the spacer is installed.

2. Installation Instructions

- 1. Raise Vehicle:** Elevate the vehicle using an appropriate lifting device, allowing the axle assembly to hang freely. Securely support the chassis before proceeding.
- 2. Remove OEM Bump Stop:** Detach and remove the vehicle's original equipment manufacturer (OEM) bump stop.
- 3. Assemble Spring and Bracket:** Integrate the hollow rubber spring with its corresponding mounting bracket. Use the spacer only when necessary for proper fitment or the recommended unloaded clearance.
- 4. Mount Assembly:** Secure the assembled helper spring unit to the vehicle frame utilizing the existing OEM fasteners.
- 5. Lower Vehicle:** Carefully lower the vehicle back to its resting position on the ground.

Installation Notes:

- When removing OEM fasteners, the application of penetrating fluid is recommended to facilitate removal.
- Confirm that all hardware is properly seated and that the spring does not interfere with nearby vehicle components.

3. Clearance Instructions & Fitting Guidance

- **Clearance:** With the vehicle unloaded and at normal ride height, maintain approximately **1/2" to 1-1/2" of clearance** between the rubber spring and its contact point. **Approximately 1" of clearance is ideal.**
- **Ride Comfort:** Proper clearance allows the factory suspension to operate normally before the helper spring engages and helps maintain a comfortable unloaded ride. Insufficient clearance or excessive preload may result in a firmer ride.
- **Load Effects:** A partial load, cargo, aftermarket accessories, suspension wear, or vehicle modifications may reduce or eliminate the clearance. Light contact under these conditions does not necessarily indicate improper installation.
- **Spacer Use:** Use the spacer only when necessary to achieve proper fitment and the recommended unloaded clearance. Use the longer bolt when the spacer is installed and the shorter bolt when the spacer is not used.

Important Safety Warnings

- Do not exceed the vehicle manufacturer's recommended axle weight rating.
- Inspect for potential interference between the hollow rubber spring and adjacent vehicle components. Under load, the rubber spring will expand in diameter.
- Support the vehicle securely before working underneath it. Follow appropriate shop safety procedures and verify the installation before driving.

Note: Illustrations are for reference only. Vehicle configuration and actual component appearance may vary. Install according to the vehicle-specific mounting points and standard mechanical practices.